

ActInf Textbook Group ~ Cohort 1 ~ Meeting 10

(Chapter 5 pt. 1)

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hello it is july 7th 2022 it is cohort one textbook group week 10 chapter five so we're having the first discussion of chapter five we'll have two weeks of discussing chapter five then in our last two weeks of july talk about more general thoughts that we have on chapters one through five and look forward to the second half of the book um [Music] here there's two new columns this first one is if you'd like to join for chapters one through five cohort two beginning in september 22 and that will be in parallel with also this first cohort continuing for chapters six through ten so just put a check mark if you want to stay in otherwise we won't like include you in that cohort okay so there's only a few questions prepared for chapter five so let's start with any general thoughts that people had about chapter five like how is this chapter similar or different from the previous ones and what is the function of chapter five here or what what did they think was an important overall aspect of this chapter and then of course anyone who can capture questions that people are asking here just add them but other than that it would be awesome to hear like what did people think chapter five was doing or what did they wonder about about the material or related to it ali and then anyone else uh i think chapter five uh needs a lot of neurobiological background and without uh necessary requisites uh it can be probably daunting chapter and unlike other chapters which are at least the previous chapters uh which concern themselves with mathematical formalisms uh here in chapter 5 for the very first time we encounter some empirical evidence for uh or the empirical uh reasons behind uh modeling active inference as such so it's a very important chapter in my opinion but at the same time without any prior neurobiological background uh it could be a really daunting chapter as well nice thanks for the summary uh mike yeah it was an interesting choice to go all the way to the human brain to illustrate how these concepts would map to biological systems as opposed to starting with more simplistic organisms and kind of working up from there yeah very nice point any other overview thoughts and then we can just like scan through it and look at the order of a few things and then continue to write questions again like as you're hearing

them or like as you're just wondering but let's um yeah hear any other general thoughts if anyone wants to raise their hand and then look at the order of the chapter well the chapter is titled message passing in neurobiology message passing was brought up in a previous chapter in this box 4.1 in a formal yet slightly non-standard notation without examples just generally the quote anyone have any thoughts on jeff yes i kind of disagree with the quote i i think uh recently we've we've learned that the plants actually uh sense and act on their their their uh on what they they detect in the environment for example uh certain trees when they're being munched on by um say like giraffes will emit uh molecules that will disperse and other trees that would that sense these uh this essentially an alert signal will start producing uh compounds in the leaves that make the leaves taste very bitter and thus discouraging uh foraging and so forth and so on and also we you know i think it's clear that plants sense and act and respond to their environment and act in order to maintain their integrity or as individuals or as uh as groups in the example i just mentioned nice agreed oh also the quote might have like a little bit of hyperbole slash comedy like two types of animals plants it's about nervous systems ali uh yeah and uh one other thing about rodolfo linus is um in one of his most famous books eye of the cortex from neurons to self um actually he describes thinking uh as an internalized movement uh and it's a very interesting way to describe uh thinking in my opinion and uh it directly relates to this active inference uh framework so yeah putting uh his quotation here might be quite relevant cool and then i added a 2017 paper with fristen and calvo about plant and predictive processing framework and then here's a like a non-active paper but about more on the behavioral side which is kind of the view from the outside without denying the view from the inside as well but it's just related to behavioral modeling okay in the uh introduction they write let us take a step back from the technical material of chapter four which as um ali and others have reminded us was like even at the beginning was like you can bypass this chapter if you don't want technical detail maybe somebody had read it even one two three five and maybe they didn't look at the appendix turn to the process theory's accompanying active inference drawing a distinction between a principle and a process theory does anyone have any thoughts on that like how as they're reading the book are they thinking about where free energy principle is related to active inference how does introducing that kind of broad map territory slash principle process distinction here help transition the book towards focusing on these empirical cases ali with a hand raise or anyone else okay just one yeah yeah go for it yeah sorry just one thing i forgot to mention about that quotation is uh as an interesting research has been emerging uh that's uh kind of so consciousness to plants as well

uh one of the articles uh one of the early articles for that is insights into plant consciousness uh so uh in fact we don't necessarily uh limit it to just human or even animal consciousness and we're beginning to relate consciousness or any other aspects of nervous system to plants as well so just one additional remarks about that previous code sorry cool yeah and the mathematical like framework for behavior and consciousness that are integrative will allow comparisons of different kinds of behavioral systems and that like leads to people wondering about consciousness and all those other areas mike and then anyone else yeah so i'm tying that back to the part of the chapter that you're on previously in terms of moving from principle to process and and you had highlighted that the latter allows us to develop hypotheses that are answerable to empirical data so i think this is um the the first part of the first instance in the book where they are making a serious effort to tie out to real world observations as opposed to saying here's a framework for how things might work and how the pieces might be connected now we're going to take that framework and think about uh can we identify things that we see in real systems in this case in the human brain and tie those back into the ideas that we put forth previously thanks totally agree it's like partitioning off the principles from the earlier chapters like in principle those two sides of the equation are equal because of how the axioms of math are or like in principle one could do this with a bayesian graph and then that is like it's like in principle you could do a linear regression with a least squares error minimization nobody can make any data set that says that you can't do that and then there's how that least squares regression works on the empirical data so they're saying for chapter five we're going to look at with all the architecture developed a few specific neural systems and talk about how they're the best current uh understanding that they have of how these neural systems can be modeled with the models of the uh structures like we've seen before and they're saying we're not trying to write a neuro textbook here but to touch kind of like with both feet the empirical modeling and then the systems that they are going to model are like effector systems so motor control subcortical structures like the thalamus and basal ganglia the which are so these are more like um functional outcomes of the nervous system specific anatomical structures of interest modulation of synaptic efficiency or efficacy which is like a micro anatomical or like a neurophysiological mechanism of interest if and something that is an important like leverage point and something that is being modified and found to be changed in different situations and then the relationship between decision making and movement generation which is also something that has come up in those um 46 live streams on like motor active inference decision making active inference so what did uh well let's check if there's any

questions written um what could anyone say about the cortical layers or what was in figure 5.1
mike so i was just observing that in the start of chapter four they said well you can skip this
chapter if you don't want to get deep into the math and then here we are at section 5.2 in
chapter 5 i think and they refer back to the message passing structure that was set up in
chapter four yes good point and it's like well if you can skip the uh details of the formalism to
look at the examples or you could skip the examples any linear text 140 characters or textbook
or you know live stream or anything it's always going to be some sort of linear presentation
question and with a densely linked and interdisciplinary area then like the linearity becomes
challenging so maybe with more you know with some notebooks or some other rendering or
educational way to have the ordering or have the of course the connections between sections
be a little bit different um the first system that they're focusing on is the cortical column does
anyone have like any thing to add on cortical layers or the role of the cortex even if it's just
something that they've heard about well i think that this um numenta line of ai and um many
of the other oh i'll leave go for it yes i just i want to say that uh well for the past and couple of
decades or probably several decades uh cortex uh has traditionally been thought of as the seat
of both consciousness intelligence and also of course cognition but recent studies has shown
that especially studies such as mark some research has shown that in fact consciousness
actually stems from uh literally the brain stem not uh necessarily the cortex so uh there are
some challenges about the exact function of cortex and the debate goes on nice okay here's a
little bit on the cortex so it is the outer frontal part of the mammalian style brain insects don't
have though they share a lot of the same architectures and so on so this is like neuron tracing
and there's kind of two levels of organization that people are modeling in the cortical models
the first is like based upon the histological observation of the layers these are six layers of like
cells and there's like a lot of development and all different functionality between the different
layers um one important thing to note is like it's not that they're like the layers of a bayesian
hierarchical model they're not like it's not like each one is like up and down up and down so
this isn't like a predictive processing architecture happening simply with these six layers this is
not a six layer model in fact they're connected to each other sparsely within a column as it's
shown here in 5.1 so that's like the micro anatomical structure of the cortical column and it's
called a column because it's like the layers are also arranged in this repeating way and there's
just like a lot of different um layouts for different regions of this cortical tissue type and in
certain situations it is organized like linear corner structures which is often seen as like one of
the evolutionary local maxima of like depth of processing and sparsity of connections laterally

so those are the two levels of analysis that are being and as ally mentioned like people yes i've gone to the consciousness angle or they've just argued that this is kind of like the massively paralyzable but also like deep recurrent contextual etc the anagrams like the patterns that can be activated the dimensionalities potentially super high so it could play a role in memory signal processing and the two levels that people look at the connections between like anatomically by taking these images or by um doing like stainings and stuff um anatomically and functionally is within the column where there's certain kinds of relationships and then across columns where there are certain kinds of relationships any thoughts or questions on that but like that's just like why the cortex is being studied and that's like a little bit of how like the neuroscientists have been approaching this and why this is one of the important systems with a lot of theoretical modeling and empirical evidence and like relevance as well um jf i remember many many years ago taking a course in physiology on the visual cortex and you know the whole notion that the brain has some kind of structure um always i thought was fascinating because i i feel that the brain is a huge hack patches upon patches upon patches brought by evolution and that there would be some structure in it is is is interesting but within within those structures there's as those this picture shows this it's very messy i would never implement a system with this kind of of wild interconnectivity i'm very curious about how much we can infer from what is really a mildly structured but highly messy architecture in terms of the the process theory that we're trying to uh to elucidate from anatomy and the functionality of the brain nice thank you mike yeah that makes me think that some of that messiness is in support of redundancy and so how do we think about redundancies in systems in the context of active inference yeah and like neural systems of course have amazing in some dimensions ability to have redundancy other times seemingly there's points of failure jeff okay okay all right so just um on the left side i'm sorry oh yeah sorry uh on the comment about redundancy i agree but it's worse than that it's it's uh every part often plays many different roles and and participate in different functions which makes it quite a tangle uh in terms of uh functional analysis of parts of the brain what they do and what not so five one is showing um a more like pure base graph with the annotation of the kinds of cells that are in the anatomical layer so this is like a visualization of the anatomy based upon the regions that it connects gets inputs and outputs from and then the micro anatomy labeled according to cell types with a similar but not exactly concordant there's not a statistical test proposed for how concordant theos are it's not a statistical argument it's not exclusive of other framings of what this anatomy is doing here are labeled with different parameters the base graph architectures that have been

explored in the previous chapters compatible with message passing implementations that other papers have addressed that aren't here these are like architectures that recapitulate some of the functional and macro structural and microstructural aspects so this is like computational neuroscience and it's using these models that actually have an underlying structure that is providing utility interesting reframings of the anatomy and as the model um statistical architecture begins to recapitulate aspects of this of the cyto architecture there's like no specific threshold where it's like oh well now it's like an argument that it's you know doing that function people get tantalized by it having some oh we made this network structured like a brain or we made this network structured like a mushroom or we did the ant colony optimization so just because like this works for what it does doesn't mean this works for what anyone wants it to do it's just in this huge space of bio-inspired model architectures in um five two they move a little bit from the consideration of the columnar architecture which are cell populations um looking now like still the superficial to deep are the six layers of the cortex but now a few of the variables are highlighted and they're relationships and so the um ascending and descending prediction error in a predictive processing framework which is like often graphically seen going up and down bottom up and top-down effects is being shown in the context of the columnar where like errors from one like let's just go to where they e's are the errors and the g's are predictions and um i guess we can find what mu is but basically like this architecture within and between could be like this being like the base variable being estimated and a hierarchical nested model of that and a hierarchical model of that so there's like a generalization statistical architecture that has compatibility with the neural architecture of the brain and it does like a pure statistical and that pure statistical function has like resemblance with some things that the brain region is believed to do so that allows this statistical model to be explaining and pre predicting and useful about the brain which is like what computational neuroscience does so that was the first one of the systems that they focus on oh yeah i guess this was um then the factor systems okay let's see if we can kind of quickly look at all the systems what did anyone see or what would they like to say about figure five three neuroanatomy associated with active inference in modulating spinal motor reflexes so five one was the columnar architecture and that was described in terms of what cell types were in it here they're going to pick up on one of those cell types which is like the computational role the expectations encoded by this cell type that's found in this layer of the motor cortex and so different regions of the cortex have different apparent functions asterisk asterisk but there's like a visual cortex in a motor cortex and so on so this is in a motor region of cortical

architecture that plays a certain biological function the expectations are subtracted from the incoming proprioceptive input in the horn of the spinal cord this is like lateral cut through a spinal cord the horns are like the butterfly parts and so there's a differencing between incoming intensity of proprioception and the brain expected expectations and then the error drives muscle activity in the direction of suppression so the motor activity is like reducing the divergence based upon the sign and the intensity of the compromise between the brain's potentially arising within like layer 5 from these bet cells and the incoming sensory information which can be modeled as like why i think μ is probably the mean of that expectation i don't know what part it was mentioned um and then here's the expectation the the difference between these two it's like a causal system flow but like a little different you're getting a 5 out of 10 pressure you're expecting a 5 out of 10 pressure the motor reflex is not engaged you're getting a 7 out of 10 pressure you're expecting 5 out of 10 there's like a negative 2 difference or you know whichever way you want to be subtracting there's like a 2 or a negative 2 difference and then that triggers like some motor neuron units to fire and then that proportionally brings into alignment with the measurement they're gonna just like that example hinged from a cortical layer and cell type here is going to be hinging on the cortex and looking to an output region which is the striatum and then they're going to do a more detailed model of a basal ganglia because it's been modeled a lot more and then also have like just a textual treatment of the thalamus there's multiple papers on dopamine and active inference and so um 2012 and then this is a really interesting of research with dopamine because dopamine is like classically described as the reward molecule more reward more dopamine lower reward lower dopamine isn't the whole meme inactive that we replace the reinforcement reward learning paradigm with a imperative that has elements that reflect pragmatic value but that isn't the um overarching imperative for policy selection and so these papers show how like in the context of a generative model with expectations and preferences and dopamine means signaling something related to the predictive processing of stimuli under a mildly optimistic world model good events and surprising events are still associated with dopamine release and things that are going worse than expected are still associated with like dopamine drops but um that actually is like a more unified way to talk about dopamine's empirical outcomes as opposed to the reward learning which has to hypothesize all of these like modules that translate different kinds of things into reward jessica yeah no i think similarly to these i don't remember where i read this but it isn't like the brain like in regards to like the dopamine comment it's like it's also the anticipation on of doing something and like

the fact that you're gonna get that dopamine kick like you get basically you get a release by anticipating something and that maybe a little bit connects also with the active inference the fact that you are anticipating something you know it's like you expect that to happen and so you get that release of the dopamine in anticipation of the event happening so i don't know much but like i'm thinking it could be connected with having like a predictive unexpected free energy or something nice good insight rohan yeah uh i'm not uh well i'm not a biologist but okay so i just wanna clarify there are other systems in the body apart from just the brain alone right so you have like the autonomous uh or what is it parasympathetic nervous system that controls your heart rate uh your respiration and other things and they and also you have stuff like adrenaline and not adrenaline uh that do produce responses and other hormones also that produce responses like uh okay for hunger or satisfaction and we we basically have figured out how to modulate these things to solve modern issues for things like obesity or something like that and there are treatments available that modify these hormones levels right so is it is this not a very limiting perspective to look at dopamine alone as the only thing and my other question would be how does that convert into say motor commands so i just increase dopamine levels without any stimulus whatsoever let's say someone's taking a drug it would you say that that does not necessarily translate to a motor command right so how does it uh do this is there's like a specialized that figures out okay so here's uh here's the previous action here's increase in dopamine and here's how we modulate to further increase dopamine where exactly do you set limits on this uh process okay the two questions were about interacting physiological systems and the second one was about dopamine means role in motor selection um blue go ahead if it's related um or if you want to address anything yeah no i just wanted to respond like the the dopamine response is like so um there are a lot of interactions that are happening in the nervous system dopamine is one of the best understood i think it's one of the ones that we've studied for a long time um and the if there's like an overload of dopamine the action can actually be non-specific and that's what we see in parkinson's right so it's like an overload of dopamine into that system and and it fries it out and then you get non-specific action in the nervous system motor action the tremors and stuff um yes good call and that like question about like um non-purposeful behavior and dopamine on the positive and on the negative like failure to engage initiate action relating it like catatonic behavior and then like hyper motor activity associated with um repetitive motions of various kinds like dopey dopamine is known to be different in people who are experiencing that differently and modified by drugs that target that like system so um but to the first question about like

pluralism with respect to physiological systems for sure this model does not exclude that in fact its flexibility is its advantage because the inputs to a brain region now we could say well at this time scale it's electrochemical and at this time scale it's going to be just um you know noradrenaline and someone says well how could you only model noradrenaline say okay let's look at the model with no adrenaline and this other one and an unmodeled component it will allow integration formally physiological mechanisms that genuinely interact and as to why people write like specific papers about specific hormone systems it's just like is chunked off in that fractal and then also one important note is like um columbo and wright have written about how like we don't need a monolithic theory of any neurotransmitter or any brain region there might be like features or components of dopaminergic systems that it just explains and predicts that they do reward taken lightly that the explanation is just of course our explanation of a biological system and also you raise like beyond the brain there's also glia and other cell types so yeah the cool thing would be a framework where we don't a priori exclude factors that are important um but just to kind of go quickly through the example and get to that motor so this is an output from some putative processing that happened in the cortex and now it's projecting into some dopaminergic regions in the subcortical area like the basal ganglia it's going to be a model involving outcomes of the difference between the preferred and the expected outcomes the sigma squiggle expected free energy evaluated for a policy and the um posterior over policy is the bold pie um the figure is like shown after a lot of the description here is where they refer to the high and low dopamine state so also like the the movie slash book awakenings is about um people who have like um atypical dopamine signaling and then they're given l-dopa and they like come and like get re-involved with life and stuff um rohan yeah so well this is a little more general but okay when so we have the marco blanket and there are some internal states and the markov blanket basically connects us to the external states right uh or samples from the external states and modulates the internal states accordingly uh okay so my question is i think if the first imperative is survival and the first imperative is to keep uh this system in homeostasis which is composed of multiple interlocking and interacting complex components right how exactly would you translate like these multiple signaling modalities with multiple hormones into maintaining homeostasis with something like free energy because some of these things are compatible entirely that like you would have to have some sort of translation mechanism right so let's say that adrenaline is jumping really high and into dangerously high levels but we only know it's dangerous because there is some system that is able to predict that okay it should not go above this level because

it's going to screw up something else in the system right right multi-scale optimization and not over optimizing one lower level parameter at the cost of some higher order parameter yeah but then you come to the curse of dimensionality so how many uh different things are you going to take up before it becomes completely useless so absolutely good question empirical yeah about what computational hardware what measurement data sets what sparsity of system graph what extent of application of heuristics how accurate you need that model to be in the empirical setting um none of it in principle um addressed by just the equations all of it related to how they're actually implemented at what scale just to quickly complete through here though this is about the depletion of dopamine is being observed um creating this akinesia failure to move and exogenous dopamine promotes impulsive behaviors of multiple different kinds this is what it looks like dopamine as again stated here is balancing modulating the balance as a neuromodulator between inferring what to do and what not to do and so these two graphs both of them are getting outcomes from the cerebral cortex of the observations expected under a policy so those could be considered like predictions about at future sensory states um and the gamma is this uncertainty that's being represented by dopamine in in a like in this uncertainty function of dopamine in the direct pathway the policies and the difference between the preferences and expectations sigma squiggle with a tilde are being evaluated as part of the expected free energy of future policies and that is resulting in the selection of policies so that could be like thinking through all the chess moves and then picking the one with the lowest expected free energy the output in the indirect pathway feeds into the prior vector e habit influences policy through this acting anatomically proposed to this other region not as not defined with a specific variable so again this is not like saying like necessarily even just what it's doing this isn't the full model the papers read more but that's like with the two pathways of action selection and it's compatible with the empirical evidence from pathology and from pharmacology and genetic studies and animals other than humans and like this is um how it could be modeled with a base graph that recapitulates some of the architecture and the function they describe the roles of some different neurotransmitters that have been modeled in active and like what kinds of phenomena those measurements have been used in models a one last important part it was in the section with the basal ganglia that was uh talking about this so they show the graph of the basal ganglia and then only in this last paragraph did they describe the thalamus so just say we're just going to do one paragraph on it but basically the thalamus has these two divisions and then they provide the um interpretation that these two divisions of the thalamus could reflect first and second order statistics on other types of senses and policies

so that's like a prediction a unique prediction that potentially modeling uh observed neural responses or the bold signal from um like the blood oxygen use in that region you'll make a better model if you fit into these categories rather than reward or rather than familiarity or any number of other criteria so just simply fitting better is not the whole substance of validating the formalisms described earlier it's part of the empirical grounding and examples of like unique predictions and explanations that are provided by active and that but that was the thalamus part and then um in this very short section 5 6 they talk about how a lot of our interfaces are continuous like sensory apparatus and motor apparatus have continuous aspects but then in the cognitive and decision-making domains there's often much more discreteness even if there's multi-scale discreteness resulting in like a very finely graded continuum of alternatives still there is um discreteness involved and this was explored a lot more fully in the 46 live streams active inference models do not contradict folk psychology where they really clarify this is motor active inference in the continuous domain this is decision making active inference in the discrete domain here's what a hybrid model looks like blue oh do i still my hand up sorry oh spin up yeah so then they just basically say um some of these models dealt with continuous phenomena like specifically the reflex arc example hashtag active inference in continuous time figure 4 3 etc and then other models presented in this are compatible with discrete alternatives like discrete variables for example policy selections if there's two model policy outcomes then this is like a discrete model or the observations could be happening through discrete time so it's like different opportunities for discreteness continuous models and variables to be integrated these architectures in summary they're outlining the points of connection between the message passing implied by the generative models of chapter four which we asked you not to read gently the neurobiology of inference action and planning and then here's going to be like a summary figure for the chapter the top row is showing some computational motifs that are plausible or compatible with different like subsets of cells and regions in cortex the bottom region are showing some extracortical structures on the left is the basal ganglia shown from like one of the previous figures so it's a subcortical structure within here is a spinal cord um cross section and one can imagine that like other structures within the brain and outside would be amenable to also this kind of graph and they're just doing like kind of linking now even the models together showing the difference in planning like on the left side the basal ganglia is getting this preference difference is connecting to the expected free energy so future policies are being evaluated policies are being evaluated on the basis of their future ability to align with preferences in this pathway here that's planning habits are when the

not the preference versus expectation um difference the outcomes themselves are going to e habits are being followed which could be seen as multi-scale habits and that's influencing policy and then here's where that kind of continuous decision and modulated handoff between more planning like and more habitual is being related to the motor implementation of the expectations of the proprioceptive loops so like i want to be sitting still and then all of your muscles in a healthy situation are like not moving versus like i want my leg again speaking like loosely slash you know with it i want type word but just like i want to lift my leg that model the expectations have to be updated to allow the leg to realize being up using a non we don't say well it's more rewarding for the leg to be up we can just say this is a computational architecture that facilitated that and inc and generalizes to like multi-step planning and then there's been multiple papers 2017 and beyond that like summarized and elaborated on different aspects of this more complex continuous architectures more complex motor discrete interfacing more complex decision making active inference of course rohan uh yeah i'm just wondering because a lot of this has to do with high level uh more abstract behaviors or kind of abstract behaviors right so is there any attempt made to apply this to something simpler like the autonomous uh nervous system because not every signal goes through to the brain right so that we just go to the spinal cord and we know this because if the spinal damage you can't sometimes people can't walk or they have trouble rebreathing and so on because so or in other animals like the cuttlefish for example that changes in the pigmentation of the skin is not dependent on brain movement uh on brain activity it's just autonomous right so it does this war has there been any attempt made to apply this to much more you know those kind of systems um well uh how simple or complex the system is kind of a it's kind of like how complexly we approach it because people have been modeling single cell um graphs if that could be seen as the simplest electrochemical decision maker as well as um morphological computing and yeah there's probably a paucity of published models autonomic nervous system active inference models so there are many opportunities because they cited a huge amount of the neural empirical work here with the goal being to summarize sort of the tip of the iceberg like there aren't models like this for every single function or region or or anything nor even if there were would it be the end of research in that area so yeah there's a lot of open space to build specific models which is exactly what section six and beyond and part two will take us to which is to go from like identifying it we can quickly find out whether it's been published or not um and then build that model with the inputs and outputs that people are seeing as relevance rowan uh yeah i'm sorry for monopolizing this question time i'm sorry so i'm just

curious if this has been applied to non-biological systems in any way because that seems to be something that we could easily do right because we have plenty of data right now okay so can we is there anything like a better vision model than a convolutional neural net for example or some sort of swarm robotics attempt using something like this or even like uh game playing or something because i would be interested to read many many of these things have been um sketched or are in progress we have our project ideas in these like last minutes there's a lot of room for building projects and there's a huge value to creative ideation of like areas where it could be applied and also value for checking the literature for where it might have been like a model could be reused like a visual foraging that was from about cultural acquisition of pattern preference was repurposed for our aunt stigma g just by adding a pheromone trace so there isn't like a cookbook on what is out there or how models could be modified we hope that with active block for ins documentation it'll be clearer and clearer for people to build generative models and compose them and explore different modules so it should just be done yeah okay thank you any last thoughts or questions on chapter 5 and then next week we'll return to the more specific questions ali and and then anyone else would like a closing thought on chapter five yeah i just want to mention um thomas pars the speech uh in the i think it was a few weeks ago in the center for cognitive neuroscience berlin which i'm putting the link here in which he explained most of these topics and some more uh in perhaps a more accessible way so if anyone wants to yes that's it if anyone wants to watch it i think uh that can help in understanding the content of this chapter nice thank you yep looks like a good one to watch and using like the same figures so that's pretty maybe we could you know annotate like the figure to say at this time step here's where it's described so those who want to um feel free to stay in this room for dot tools otherwise thanks for joining and see you next week